fast approximate metric is used: this gives a less precise result, that can however be refined by using a proper clustering algorithm as a next step: as we'll see in the next section, using canopy clustering to bootstrap other algorithms can both speed up convergence and reduce the running time[4].

Figure 13.2 shows how canopy clustering works through an example, while listing 13.1 describes its pseudo-code.

Listing 13.1 Canopy Clustering

```
function canopyClustering(points, T1,
T2)                                          #1
  throw-if T1 <= T2
                              #2
  centroids ←
points
#3
  canopies ←
[]
  #4
  while not centroids.isEmpty()
do                                                  #5
    p ← centroids.drawRandomElement()
                                     #6
    canopy ←
{p}
  #7
    for q in points do
                              #8
      dist ← distance(p,
q)                                                        #9
      if dist < T1
then                                                            #10
```